Do Technical Trading Strategies Generalize Across Assets?

An Empirical Study Using Cross-Asset Backtesting

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Abstract

This paper investigates whether commonly used technical trading strategies generalize across different financial assets. Using a systematic backtesting framework, strategies including moving average crossover, RSI-based mean reversion, and Bollinger Band mean reversion are evaluated on multiple assets. A key focus is placed on cross-asset transfer: parameters are optimized on a single training asset and then applied to other assets without modification. Performance is evaluated using risk-adjusted metrics such as Sharpe ratio, drawdown, and volatility. The results indicate that while certain strategies exhibit strong in-sample performance, their effectiveness does not consistently transfer across assets. In contrast, simpler fixed-parameter strategies demonstrate more stable behavior across markets. These findings highlight the importance of robustness and generalization in quantitative strategy design.

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1 Introduction

Quantitative trading strategies use systematic rules to generate trading signals based on historical price data. Among the most widely used approaches are technical indicators such as moving averages, the relative strength index (RSI), and Bollinger Bands. While these strategies are often evaluated based on historical profitability, their real-world applicability depends on their ability to generalize beyond the specific data on which they were developed.

A major challenge in quantitative research is overfitting, where a strategy performs well on a particular dataset but fails to maintain performance when applied to new data. While this issue is commonly addressed through time-based train-test splits, less attention is given to cross-asset generalization — whether a strategy developed on one asset performs well on others.

This paper addresses this question by evaluating the cross-asset transferability of technical trading strategies. Strategy parameters are optimized on a single asset and then applied to other assets without re-optimization. This approach is analogous to out-of-distribution testing in machine learning, where models are evaluated on data drawn from a different distribution than the training set.

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2 Data and Methodology

2.1 Data

Historical daily price data is obtained using Yahoo Finance for the following assets:

- SPY (S&P 500 ETF)
- QQQ (Nasdaq-100 ETF)
- AAPL (Apple Inc. equity)

The dataset includes daily closing prices over a rolling multi-year period.

2.2 Train-Test Framework

For each asset, the dataset is divided into:

- 70% training data
- 30% testing data

However, parameter optimization is performed **only on SPY training data**. The learned parameters are then applied unchanged to all evaluation assets, including SPY, QQQ, and AAPL.

This design allows for evaluation of both:

- time-based generalization (train vs test)
- cross-asset generalization (SPY → QQQ, AAPL)

2.3 Strategies Evaluated

Moving Average Crossover (Trend-Following)
A long position is taken when a short-term moving average exceeds a long-term moving average, and exited otherwise. Parameters (short and long windows) are optimized using training data.

RSI Mean Reversion
A long position is taken when RSI falls below an oversold threshold (30) and exited when RSI rises above 55. Parameters are fixed across all assets.

Bollinger Band Mean Reversion
A long position is taken when price falls below the lower Bollinger Band and exited when price reverts to the mean. Parameters are fixed.

Buy and Hold
A passive benchmark strategy with continuous exposure.

2.4 Performance Metrics

Performance is evaluated using the following metrics:

- Total Return
- Annualized Return
- Annualized Volatility
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Calmar Ratio

Trade-level metrics include:

- Number of Trades
- Win Rate
- Average Trade Return
- Average Holding Period

2.5 Formal Problem Definition

We define a trading strategy as a function:

$S: P_t \rightarrow a_t$

where P_t represents historical price data up to time t, and $a_t \in \{0,1\}$ represents the trading action (long or flat).

Let θ denote the parameter set of a strategy (e.g., MA window lengths).

Define performance as:

$J(S, D)$ = Sharpe ratio of strategy S evaluated on dataset D.

The core problem studied in this paper is:

Does:

$$\text{argmax}_{\theta}\, J(S_{\theta}, D_{\text{train}})$$

generalize to:

$$J(S_{\theta}, D_{\text{test_other_assets}})?$$

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3 Results

3.1 Cross-Asset Performance Comparison

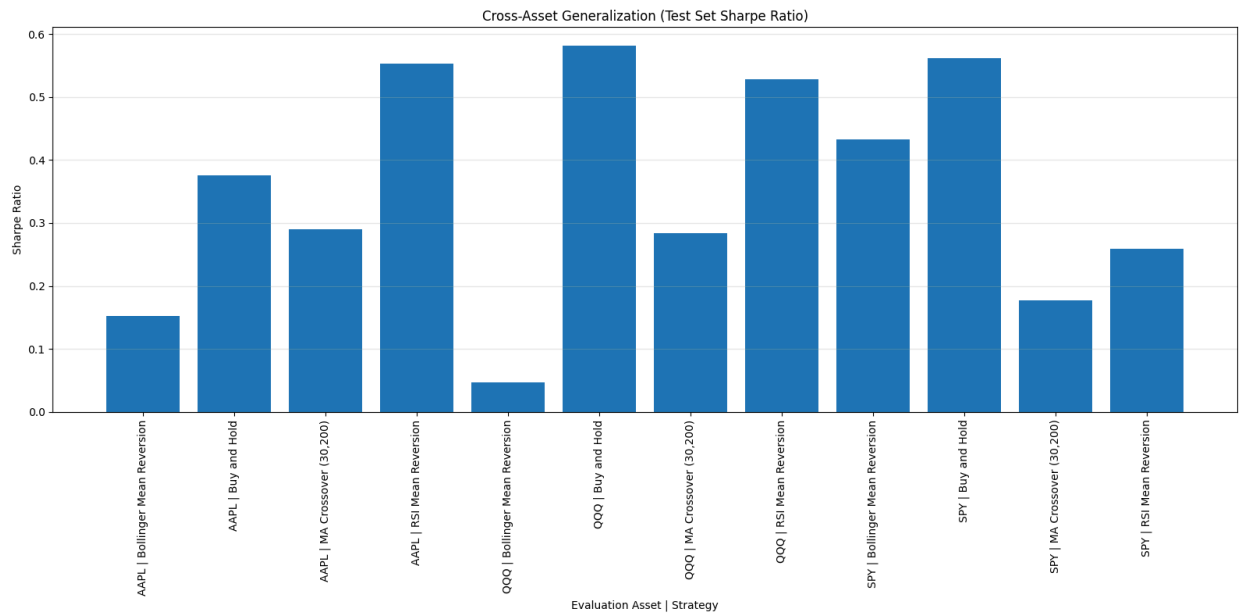

Figure 1. Sharpe ratio comparison across strategies and assets under cross-asset evaluation.

3.2 Total Return Across Assets

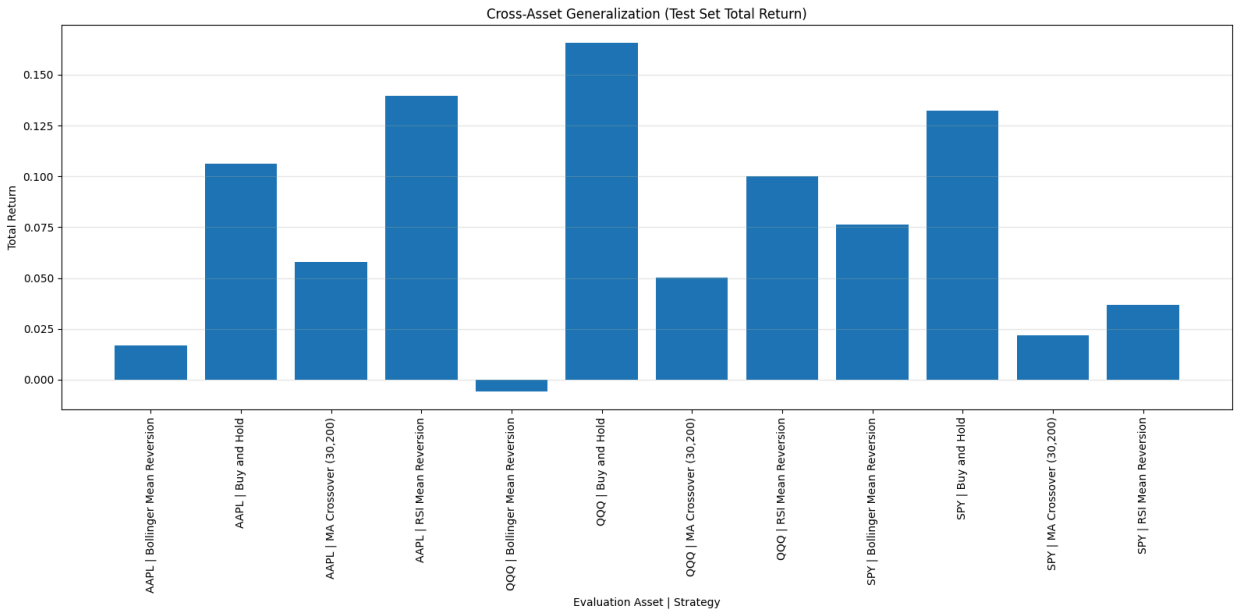

Figure 2. Total return of each strategy when trained on SPY and evaluated across SPY, QQQ, and AAPL.

3.3 Parameter Sensitivity of MA Strategy

Figure 3. Heatmap of Sharpe ratio across moving average parameter combinations.

3.4 Cross-Asset Performance of Strategies

The moving average crossover strategy, optimized on SPY (30, 200), demonstrates strong performance on the SPY training set. However, its performance deteriorates on the SPY test set and shows inconsistent results when applied to QQQ and AAPL.

This indicates that parameters optimized on one asset do not reliably generalize to others.

Mean reversion strategies (RSI and Bollinger) exhibit more stable performance across assets. While their returns are not consistently superior, their behavior is less sensitive to parameter tuning and asset-specific characteristics.

The Buy and Hold strategy performs consistently well across all assets and remains difficult to outperform. In many cases, active strategies fail to deliver significantly higher risk-adjusted returns.

3.5 Summary Table

Training Asset	Evaluation Asset	Strategy	Sample	Sharpe Ratio	Total Return
SPY	SPY	MA Crossover (30,200)	Test	0.1770	0.0217
SPY	QQQ	MA Crossover (30,200)	Test	0.2835	0.0501
SPY	AAPL	MA Crossover (30,200)	Test	0.2902	0.0579
SPY	SPY	RSI Mean Reversion	Test	0.2585	0.0367
SPY	QQQ	RSI Mean Reversion	Test	0.5283	0.1001
SPY	AAPL	RSI Mean Reversion	Test	0.5531	0.1397

Table 1. Out-of-sample cross-asset performance comparison.

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4 Analysis

The empirical results provide several insights into the behavior of technical trading strategies under cross-asset evaluation.

Limited Generalization of Optimized Strategies

The moving average crossover strategy, optimized on SPY, demonstrates a noticeable decline in performance when applied to out-of-sample data and other assets. This suggests that optimal parameter choices are highly dependent on asset-specific characteristics.

Evidence of Overfitting

The degradation in performance between training and testing samples indicates that parameter optimization may capture noise rather than persistent market structure. This highlights the risk of relying solely on in-sample optimization and indicates that parameters optimized on one asset do not reliably generalize.

Empirical evidence suggests that:

$J(S_{\theta^*}, SPY_{train}) >> J(S_{\theta^*}, QQQ_{test})$

Robustness of Mean-Reversion Strategies

Mean-reversion strategies (RSI and Bollinger Bands), which rely on fixed parameters, exhibit more stable performance across assets. While their returns are not always superior, they demonstrate greater consistency and robustness.

Benchmark Strength and Market Efficiency

The Buy and Hold strategy consistently performs well across all assets, reinforcing the difficulty of outperforming passive exposure using simple technical rules. This aligns with broader evidence supporting market efficiency in liquid equity markets.

Interpretation as Overfitting

Let $\theta^* = \text{argmax}_\theta J(S_\theta, D_{train})$

Then θ^* may capture noise specific to the training dataset D_{train} rather than persistent structure shared across assets.

Thus, in expectation:

$E[J(S_{\theta^*}, D_{new})] < E[J(S_{\theta_general}, D_{new})]$

This suggests that financial markets exhibit asset-specific structure, limiting the transferability of optimized strategies across assets and reinforcing the importance of robustness over optimization. In particular, strategies that rely heavily on parameter tuning may capture patterns that are unique to a specific dataset rather than reflecting persistent, generalizable market dynamics. As a result, approaches that prioritize stability, simplicity, and consistency across different environments may be more reliable in practice than those that achieve strong performance through extensive in-sample optimization.

5 Limitations

This study does not account for transaction costs, slippage, or market impact, which may significantly affect real-world performance. Additionally, only a limited set of assets and strategies are considered. The use of daily data may also obscure intraday dynamics relevant to certain trading strategies.

6 Conclusion

This study demonstrates that technical trading strategies optimized on a single asset do not reliably generalize across different assets. While optimization can improve in-sample performance, it often leads to reduced robustness when applied out-of-sample or across assets.

Simpler strategies with fixed parameters exhibit more consistent behavior, although their performance remains mixed. The persistent strength of passive benchmarks further highlights the challenges of developing reliable active trading strategies.

Future work may explore multi-asset optimization, portfolio-level strategies, and machine learning approaches to improve generalization.